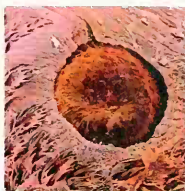


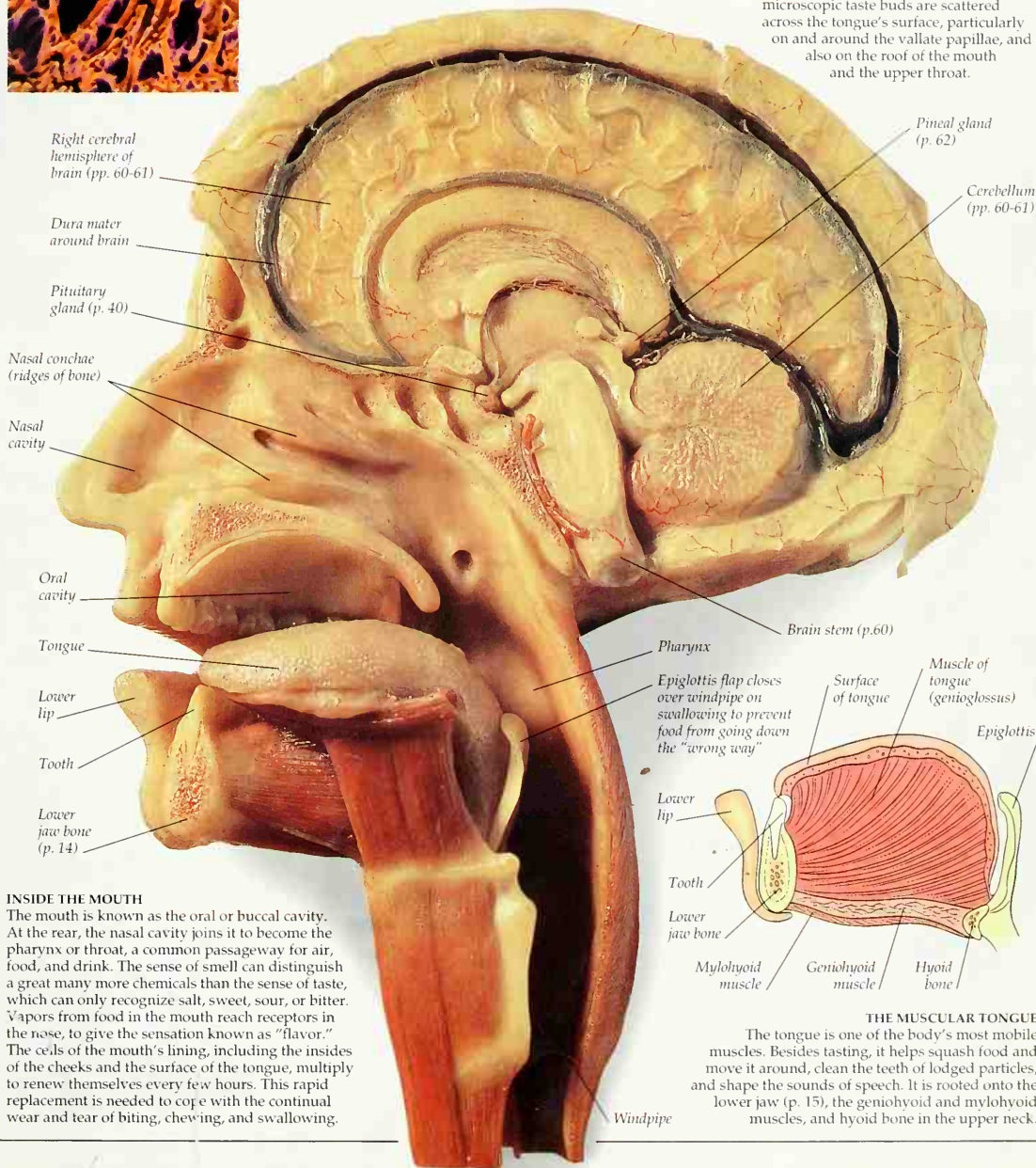
THE SMELL SENSORS

Under the scanning electron microscope, the olfactory epithelium looks like a mass of hairs and strands. These hairs are called cilia, and they radiate from bulges (one is seen, center) on the cytoplasm of the smell receptor cells (pp.12-13). The cilia bear tiny studs that are thought to be the points of interaction between the smelly substance and the nerve receptor cell. The odor molecules are believed to fit like keys in locks on the cilia, to generate nerve signals.



PAPILLAE AND TASTE BUDS

The tongue's surface is roughened by papillae, or "pimples," of several shapes and sizes. This is an enlarged view of a large, rounded vallate papilla, found at the rear; along the back sides are ridged foliate papillae; on the main upper surface are fungiform papillae shaped like mushrooms; and around the tip are hairlike filiform papillae. About 10,000 microscopic taste buds are scattered across the tongue's surface, particularly on and around the vallate papillae, and also on the roof of the mouth and the upper throat.



INSIDE THE MOUTH

The mouth is known as the oral or buccal cavity. At the rear, the nasal cavity joins it to become the pharynx or throat, a common passageway for air, food, and drink. The sense of smell can distinguish a great many more chemicals than the sense of taste, which can only recognize salt, sweet, sour, or bitter. Vapors from food in the mouth reach receptors in the nose, to give the sensation known as "flavor." The cells of the mouth's lining, including the insides of the cheeks and the surface of the tongue, multiply to renew themselves every few hours. This rapid replacement is needed to cope with the continual wear and tear of biting, chewing, and swallowing.

THE MUSCULAR TONGUE

The tongue is one of the body's most mobile muscles. Besides tasting, it helps squash food and move it around, clean the teeth of lodged particles, and shape the sounds of speech. It is rooted onto the lower jaw (p. 15), the geniohyoid and mylohyoid muscles, and hyoid bone in the upper neck.